

Read mapping

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Lecture outline

- Illumina sequencing: overview of technology
- The FASTA file format – for reference genomes
- The FASTQ file format – for raw sequencing data
- Processing short read data: read mapping vs de-novo assembly
- Read mapping – conceptual overview
- Key concepts: sequencing coverage, mapping quality
- The SAM/BAM/CRAM format – for reads aligned to a reference genome
- samtools: your best friend when processing sequencing data

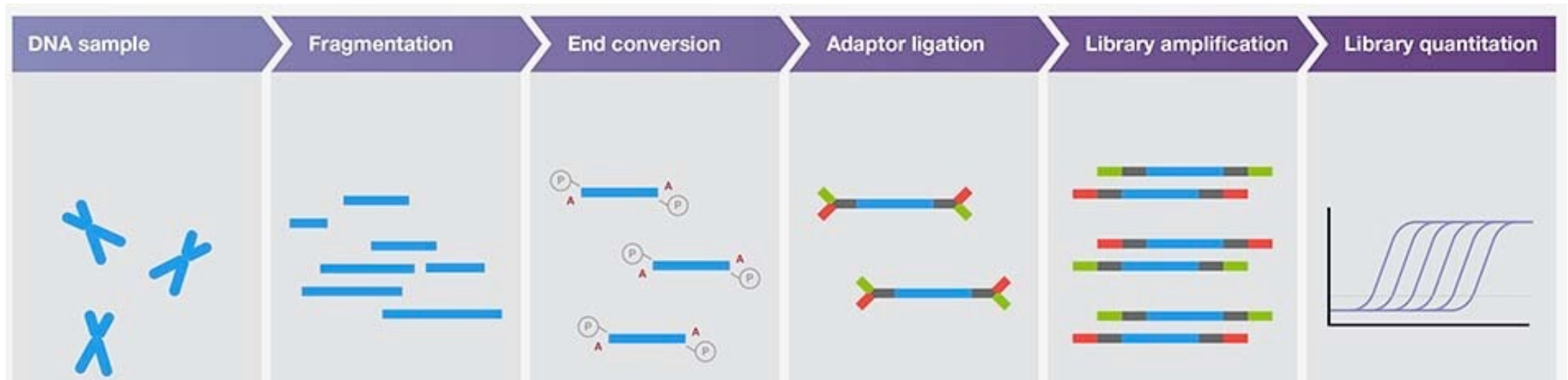
Illumina sequencing

- High throughput
- Short reads: Usually 100-150 base-pairs
- Scalability
- Lower cost
- MiSeq
- HiSeq
- Novaseq6000



Library preparation: preparing DNA for sequencing

- PCR-Free
- Amplicon sequencing
- Pooled samples – use of barcodes
- Strandedness

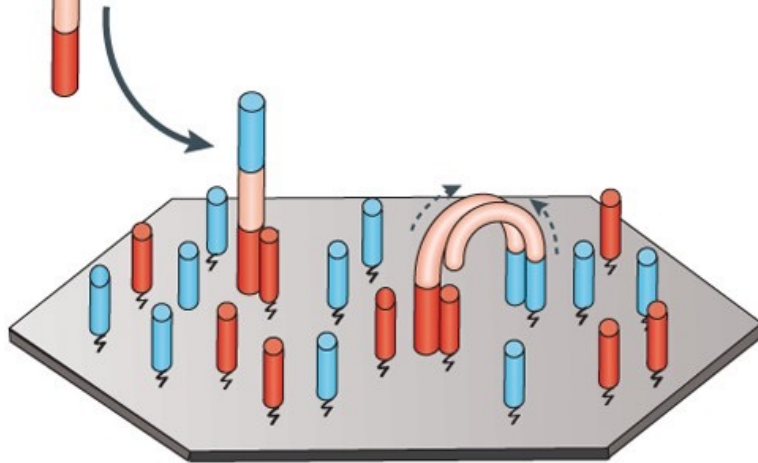


The Illumina flow cell: where sequencing happens

b Solid-phase bridge amplification (Illumina)

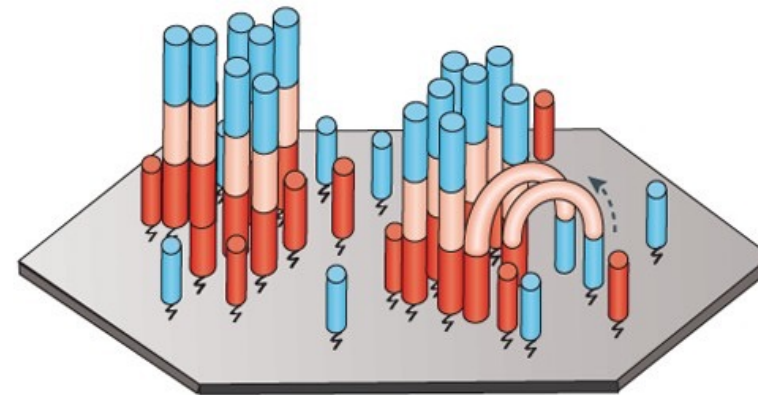
Template binding

Free templates hybridize with slide-bound adapters



Bridge amplification

Distal ends of hybridized templates interact with nearby primers where amplification can take place



Cluster generation

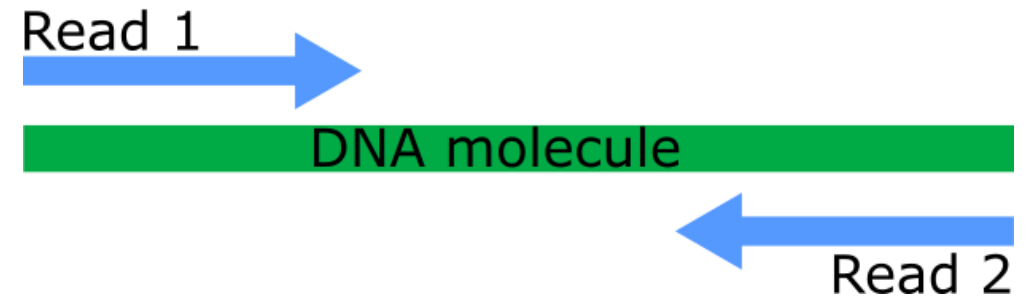
After several rounds of amplification, 100–200 million clonal clusters are formed

Single vs Paired-end sequencing

Single-end:



Paired-end:



Reads	Pro's	Con's	Application
Single-end	Lower cost	Lower accuracy in mapping	Small-RNA-seq Amplicon seq
Paired-end	Better mapping accuracy	Higher Cost More data to analyse Longer turnaround to generate data	WGS RNA-seq Metagenomics

Different sequencing strategies

	Pro's	Con's	Use case
Short read	More pipelines Lower cost	Less accurate for repetitive regions and structural variation	Standard Illumina SNPs, RNA-seq, DNA-seq
Long read	Higher accuracy for repetitive regions Best for detection of structural variation	Higher cost Higher error rate	De novo genome assembly Isoform analysis PacBio Oxford Nanopore
PCR-free	Reduced bias Fewer errors More uniform coverage	Higher cost Higher sample input needed Higher coverage harder to achieve	Clinical diagnostics CNV studies
Pooled	Low cost High-throughput	Some extra analysis steps Risk of contamination	Amplicon sequencing Metagenomics

PCR duplicates

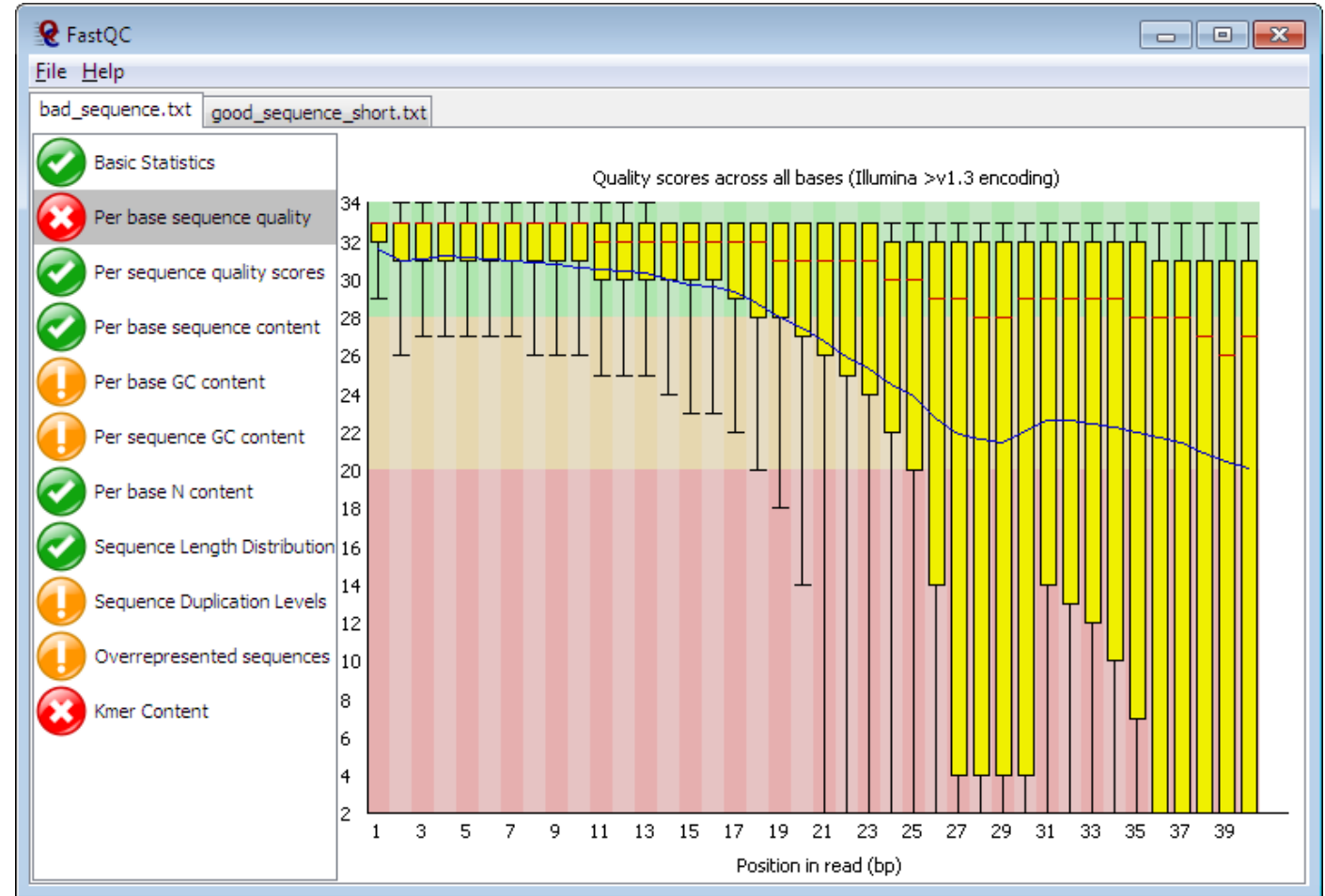
- Duplicate reads created by PCR during library preparation
- Can skew abundance of sequences
- Can reduce diversity in data
- Can inflate read depth – false-positives

Detection:

- Paired end reads
- Use of UMI tags
- Picard MarkDuplicates
- Optical duplicates

Base Quality scores

- Phred scores
- Q score
- MultiQC
- Adapter trimming
- Low quality reads
- uncalled bases



FastQC: a software for FASTQ quality control

The FASTA format

The FASTA format is the standard format for storing DNA (or RNA, or protein) sequences.

The name was inherited from an early software package called FASTA. It is typically pronounced "*fasta*" or "*fast A*"

```
>sequence identifier  
sequence
```

Example:

```
>chr1  
NNNNNGAGGGCTGAAGAAATGGTAGAACGGAGCAGCTGGTGATGTGTGTTGGGCCCACCGGCCCCAGGCTCCT  
GTCTCCCCCCCAGGTGTGTGGTGATGCCAGGCATGCCCTTCCCCAGCATCAGGTCTCCAGAGCTGCAGAAG  
ACGACGGCCGACTTGGATCACACTCTTGTGAGTGTCCCCAGTGTTGCAGAGGTGAGAGGAGAGTAGACAG  
TGAGTGGGAGTGGCGTCGCCCCCTAGGGCTCTACGGGGCCGGCGTCTCCTGTCTCCTGGAGAGGCTTCGAT  
GCCCCTCCACACCCTCTTGATCTTCCCTGTGATGTCATCTGGAGCCCTGCTGCTTGCGGTGGCCTATAAA
```

The FASTQ format

The FASTQ format is an extension of the FASTA format, which allows includes quality scores for each base.

@read name

sequence

+ (this third line is meaningless, only there for historical reasons)

base qualities

Example:

```
@LH00235:244:227WHLLT4:8:1432:43023:25836 1:N:0:GAACCTCGAT+CAAGACGCTT
AAATCAAGGTAGCTACTGCTTAATCTCTACCTTAGAAGAGCACAGAAGAGAGCATGGACTGTGTGCTCTGC
+
!' '*((( (***) ) %%%++) (%%%) .1***-+*' ')) **55CCF>>>>>CCCCCCC65)) **>CCCCC
```

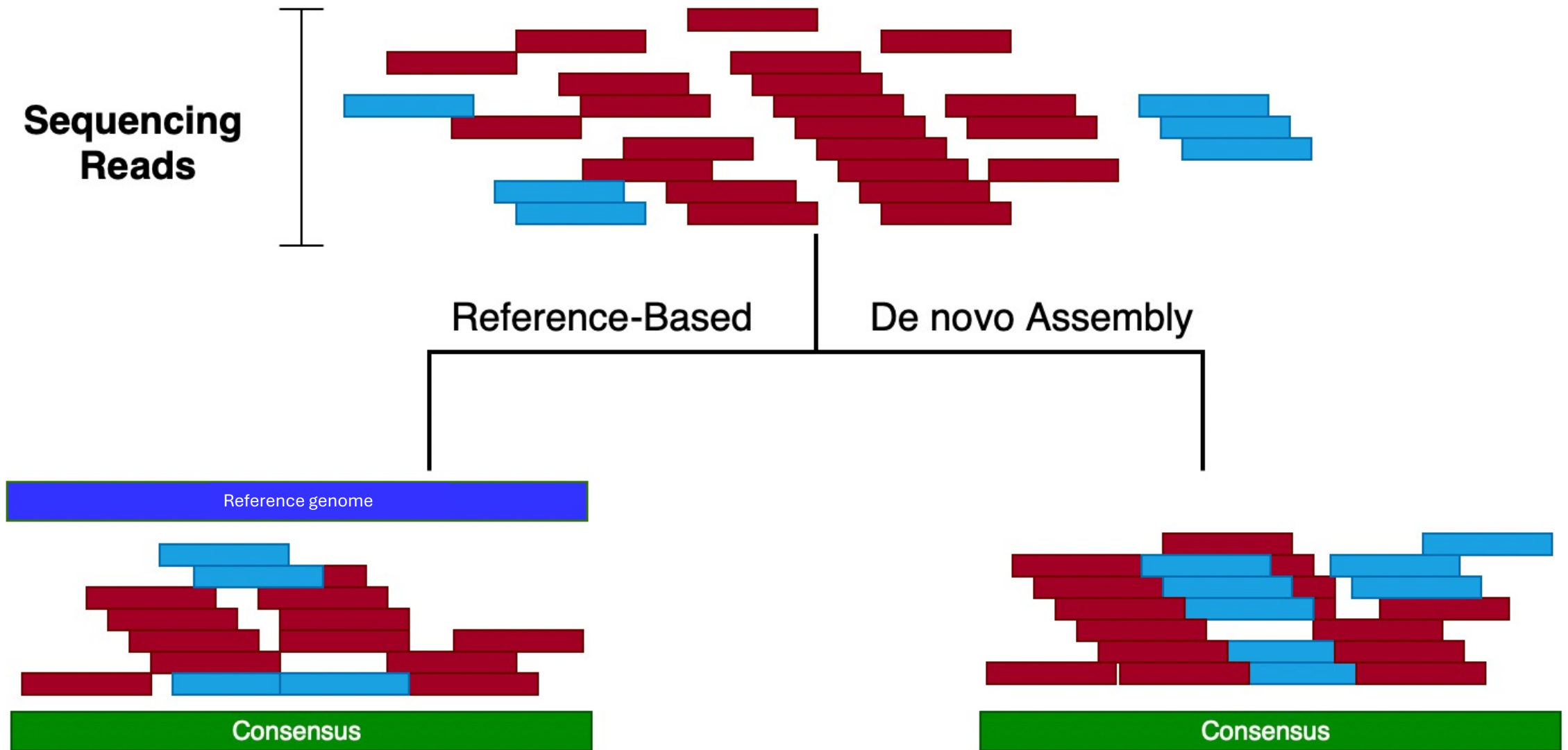
Paired-end FASTQ files

With data from a paired-end sequencing experiment, we get two reads from each molecule. These will be present in two separate files, typically ending in ‘_1.fastq.gz’ and ‘_2.fastq.gz’. The two reads in a pair are sometimes called ”mates”.

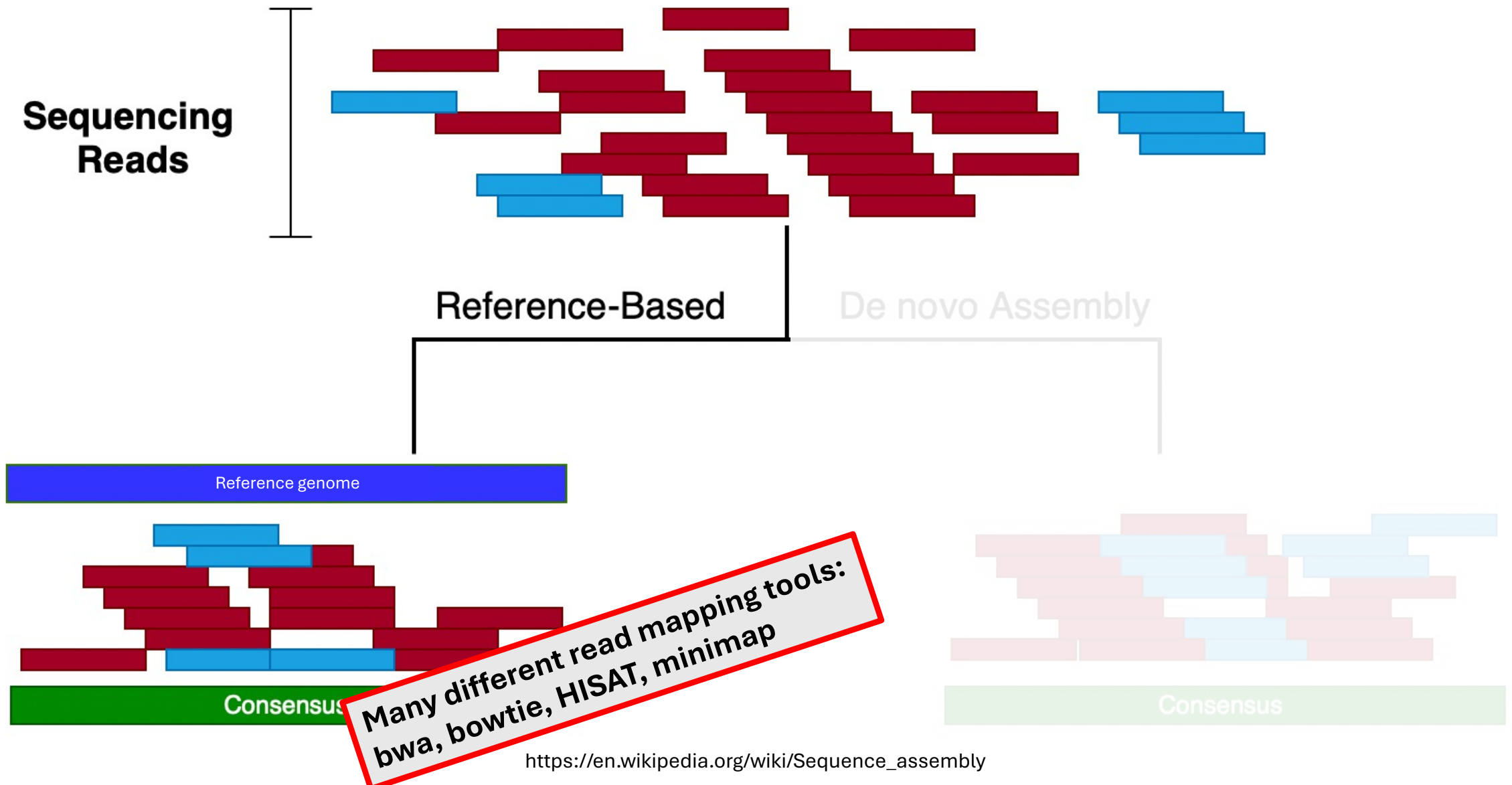
The two reads in a pair are on the same line in the files – so line order is crucial!

experiment_1.fastq		experiment_2.fastq
@read1/1	←→	@read1/2
AAATCAAGGTA		AGAAGAGCACA
+		+
! ' * ((((***		* ' ') **55CC
@read2/1	←→	@read2/2
GCTACTGCTTA		GAAGAGAGCAT
+		+
+)) % % % + +) (%		F > > > > > CCCC
@read3/1	←→	@read3/2
ATCTCTACCTT		GGACTGTGTGCTCTGC
+		+
% % %) . 1 * * * - +		CCC65)) * * >

Read mapping vs. de-novo assembly



Read mapping vs. de-novo assembly



Reads in a FASTQ file



A reference genome



Reads in a FASTQ file



A reference genome



Reads in a FASTQ file



A reference genome

Reads in a FASTQ file



A reference genome

(Unmapped reads)



Reads in a FASTQ file



A reference genome

(Unmapped reads)



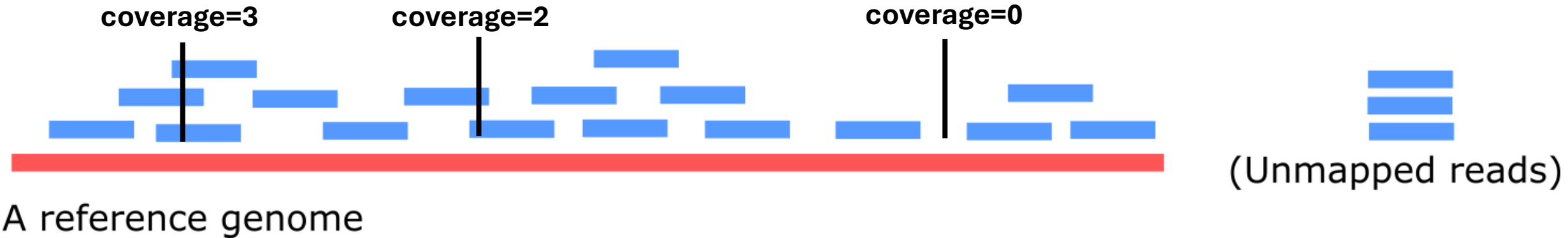
Reads in a FASTQ file



Sequencing coverage

An important concept is "**coverage**", also known as "**depth**". It refers to the number of reads that are mapped to a given location in the genome.

It can also be averaged across many regions, or the whole genome: "**genome-wide coverage**" / "**genome-wide depth**". Often, the shorthand "x" is used to refer to coverage. For example, "*the genome was sequenced to 8x*" means that on average, each position in the genome is covered by 8 reads.



In genome sequencing experiments, >20x coverage might be considered "high-coverage", whereas <5x might be considered "low-coverage". Note that genome-wide average coverage can be below 0, e.g. 0.1x.

Mapping quality

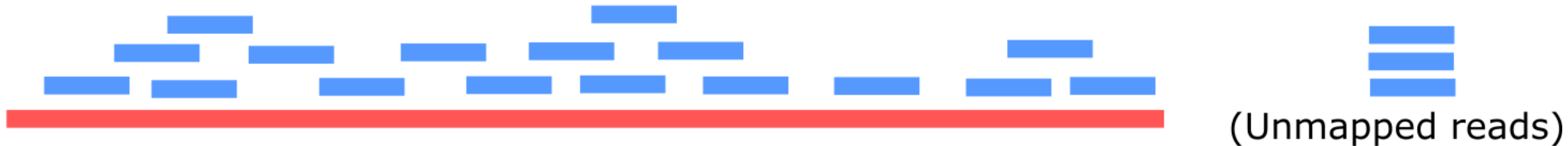
The mapping software assigns each read a “mapping quality” (often “MAPQ” or “MQ”). This represents the confidence of the software that the read has been mapped in the correct place. Importantly, it’s on a logarithmic scale.

Mapping quality

10
20
30
40

Probability of being mapped incorrectly

1 in 10
1 in 100
1 in 1000
1 in 10000



A reference genome

If a read can be aligned equally well in more than one place in the reference genome, it gets a mapping quality of 0. The software will pick one of the places at random, but it means we can not be sure if it’s the right place.

The SAM/BAM format

Example of a SAM file. Header lines start with “@”, and are followed by the read alignment lines.

```
@HD      VN:1.6   SO:coordinate
@SQ      SN:chr1  LN:248956422
@SQ      SN:chr2  LN:242193529
@SQ      SN:chr3  LN:198295559
@SQ      SN:chr4  LN:190214555
@SQ      SN:chr5  LN:181538259
@SQ      SN:chr6  LN:170805979
@SQ      SN:chr7  LN:159345973
@SQ      SN:chr8  LN:145138636
@SQ      SN:chr9  LN:138394717
...
@SQ      SN:chr21   LN:46709983
@SQ      SN:chr22   LN:50818468
@SQ      SN:chrX    LN:156040895
@SQ      SN:chrY    LN:57227415
@SQ      SN:chrM    LN:16569
@PG      ID:bwa     PN:bwa     VN:0.7.17-r1188 CL:bwa mem /gpfs/data/BIO-DSB/Session5/ref/human-ref-GRCh38.fasta /gpfs/data/BIO-DSB/Session5/datasets/datasetA_1...
@PG      ID:samtools PN:samtools PP:bwa  VN:1.21 CL:samtools view -b -o out/datasetA.bam -
@PG      ID:samtools.1 PN:samtools PP:samtools VN:1.17 CL:samtools sort out/datasetA.bam
@PG      ID:samtools.2 PN:samtools PP:samtools.1 VN:1.17 CL:samtools view -h datasetA.bam
SRR043348.10645170 163 chr1 82822 0 76M = 83175 429 ACCTTGAAGAAGGGAAAAATTTGGTTTCCAGAATAAACACATTATAATATCCAAAATGTCCAGTTT...
SRR043348.10645170 83 chr1 83175 0 76M = 82822 -429 AAAAAAGAATCAGTGAACTTGATAGATCAAATGAAATGATTTCAGTCTGAAGAGCAGGAAAAATGAAA...
SRR043354.10635865 163 chr1 90073 27 76M = 90430 433 AGTCCCTCAGTCCCTCTGTCTCTGCCAATCAGTTAACCTGCTGCTTCCTGGAGGAAGACAGTCCCT...
SRR043354.10635865 83 chr1 90430 27 76M = 90073 -433 CACTCTGTCTCTGCCAACCCAGTTGACCGCAGACATGCAGGTCTGCTCAGGTAAGACCAGCACAGT...
SRR043348.9867649 163 chr1 97700 0 76M = 97979 355 TTTAAGAGCTTCCTCTCCAGTTGAGCATGTAAGAAAAATTATACCAGGAGAATACAGTAAACTCTAT...
SRR043378.6983361 99 chr1 134158 19 51M1D25M = 134485 403 CCTCTTCCCGCTGGGTTCACGCCATTCTCCTGCCTCAGCCTTCCGGGTAGCTAGGATTACAGGTGG...
SRR043378.6983361 147 chr1 134485 19 76M = 134158 -403 AGAGCTTTTCACTTTTCGAAAAATAGCTCCTGCTGGGGCTACAAGATGGAGTGTGAAGAGGGCCTT...
SRR043372.721651 161 chr1 149796 0 76M chr19 215194 0 CACCCATCTTGGATAAAAAATGCAACAACACAATTTTCATTTTGTTCATTAACAAAACCGATTAAAGTAG...
SRR043348.5144599 99 chr1 153212 0 76M = 153557 421 CCAGTGATTTTATCAATCATGCCTCCATAAACACCCAAACAGCAGGGTTTGGAGAGCTTCTGTGTT...
SRR043348.5144599 147 chr1 153557 0 76M = 153212 -421 AAAAGAAAAGTAGCCAAAAGTGGTGGTGGGAACCCCTGTAATTCCAGCTACTTGAGAGGCTGAAGCAG...
SRR043372.8930544 129 chr1 167534 0 76M chr16 9013484 0 TGGGAGGCTGAGGTGAGTGGATCAGGAGGTGAGCAGTTCAAGACCAGCCTAGCCAAGA...
```


SAMtools

SAMtools is the primary software used to work with SAM/BAM/CRAM files. It can be used to perform a range of common operations on these files.

Command	Description
<code>samtools view</code>	View files, convert between SAM/BAM/CRAM, filter reads
<code>samtools index</code>	Build an index for a file (".bai"), to allow for faster data retrieval
<code>samtools sort</code>	Sort reads in a file by chromosomal position, or by read name
<code>samtools merge</code>	Merge multiple files into one
<code>samtools split</code>	Split a file into multiple files
<code>samtools idxstats</code>	Retrieve index statistics: number of reads mapping to each chromosome
<code>samtools flagstats</code>	Retrieve flag statistics: number of reads with different mapping properties
<code>samtools depth</code>	Calculate number of reads ("depth") at each position of the genome
<code>samtools markdup</code>	Identify and mark duplicate reads
<code>samtools fastq</code>	Convert a BAM file to a FASTQ file

Examples

<code>samtools sort myreads.unsorted.bam > myreads.sorted.bam</code>	<code>#sort reads by chromosomal position</code>
<code>samtools view myreads.bam chr3 > myreads.chr3.bam</code>	<code>#extract reads from chr3 only</code>
<code>samtools view -q 30 myreads.bam > myreads.MQ30.bam</code>	<code>#extract reads with mapping quality ≥30</code>
<code>samtools merge -o merged.bam chr1.bam chr3.bam chr4.bam ...</code>	<code>#merge separate BAMs into one</code>